



An evolutionary algorithm based on fully connected weight networks for mixed-variable multi-objective optimization

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ABSTRACT

Multi-objective optimization research has mostly focused on continuous-variable problems. However, real-world optimization problems often involve multiple types of variables (continuous, integer, and discrete) and multiple conflicting optimization objectives, called mixed-variable multi-objective optimization problems (MVMOPs). Discrete variables make the decision space of the problem discrete. In contrast, while different types of variables need different treatments by the evolutionary algorithm, which poses a challenge to the efficient search of the evolutionary algorithm. Therefore, we propose an evolutionary algorithm based on a fully connected weight network (FCWNEA). The fully connected network structure characterizes the entire decision space, the node access count records the frequency of visits to the node, and the weights of connections and the activity of variables estimate the distribution of the decision space. This information assists in generating offspring solutions. To evaluate the performance of the proposed algorithm, we conduct empirical experiments on different types of problems. The results show that the proposed algorithm has a significant advantage in mixed-variable multi-objective problems. Moreover, the proposed algorithm is also quite competitive in continuous problems and can better handle the correlation between variables in optimization problems.

1. Introduction

Multi-objective optimization problems (MOPs) and mixed-variable optimization problems (MVOPs) widely exist in the real world, such as chemical processes, electronic circuits, and communications. However, MOPs and MVOPs often do not exist independently of each other. The optimization problems in various fields can be mixed-variable and multi-objective, called MVMOPs. For example, in a hybrid renewable energy system (HRES) [1], in which the photovoltaic installation angle and the wind tower height are continuous variables, the number of components (photovoltaic panel, wind turbines, diesel, and battery) are integer variables, and the type of components (photovoltaic panel, wind turbines, diesel, and battery) are categorical variables. The optimization objectives include minimization of the loss of power supply probability, minimization of annualized system cost, minimization of greenhouse gas emission, and so on.

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In general, the main objective of mixed-variable multi-objective optimization is to determine the various variables to obtain a nondominant solution set subject to some constraints. Let n_r , n_i , and n_c be the number of continuous, integer, and categorical variables, respectively. The total number of variables is $n = n_r + n_i + n_c$. Then, the mixed-variable multi-objective optimization problem may be expressed in mathematical form as follows:

$$\text{Min } F(X) = (f_1(X), f_2(X), \dots) \quad (1)$$

subject to

$$h_i(X) = 0 \quad i = 1, 2, \dots, m$$

$$g_j(X) \leq 0 \quad j = 1, 2, \dots, n$$

$$X = [X^r, X^i, X^c] = [x_1^r, \dots, x_{n_r}^r, x_1^i, \dots, x_{n_i}^i, x_1^c, \dots, x_{n_c}^c]$$

$$x_i^{rl} \leq x_i^r \leq x_i^{ru} \quad \text{for } i = 1 \dots n_r$$

$$x_i^{il} \leq x_i^i \leq x_i^{iu} \quad \text{for } i = 1 \dots n_i$$

$$x_i^{cl} \leq x_i^c \leq x_i^{cu} \quad \text{for } i = 1 \dots n_c$$

where $h_j(X)$ and $g_j(X)$ denote equality and inequality constraints, m and n are the number of equality and inequality constraints, respectively. $X^r \in \Omega^r$, $X^i \in \Omega^i$, and $X^c \in \Omega^c$ denote a feasible subset of continuous, integer, and categorical variables, respectively. x_i^{rl} , x_i^{il} , and x_i^{cl} are the lower bounds of the i th continuous variable, integer variable, and categorical variable, respectively. x_i^{ru} , x_i^{iu} , and x_i^{cu} are the upper bounds of the i -th continuous variable, integer variable, and categorical variable, respectively.

MVMOPs have three basic features:

- As a MOP, the optimization objectives conflict with each other, and no single solution can optimize all the objectives at the same time. A pareto set (PS) sets all the optimal tradeoffs in the decision/objective space.
- As an MVOP, the integer and categorical variables divide the decision space Ω into several discontinuous feasible parts.
- The optimal Pareto set may be distributed in different subspaces.

The combination of these three features increases the difficulty of optimizing the problems.

Multi objective optimization problems (MOPs) have been widely investigated, and evolutionary algorithms have emerged as a powerful tool for solving them. Some examples of such algorithms are nondominated sorting genetic algorithm II (NSGA-II) [2], indicator-based evolutionary algorithm (IBEA) [3], and multi-objective evolutionary algorithm based on decomposition (MOEA/D) [4]. However, most of the existing MOP research has focused on continuous multiobjective optimization, and developed various reproduction operators, such as binary crossover, simulated binary crossover (SBX) [5], polynomial mutation (PM) [6], differential evolution (DE) mutation [7], and particle updated operator [8]. These reproduction operators are not directly applicable to integer or categorical variables.

Several classical methods have been proposed to tackle MVOP problems, such as the simplex method [9], cutting plane method [10], branch and bound method [11], and implicit enumeration method [12]. However, these conventional methods often assume that the problem model is differentiable and continuous, which limits their applicability to nonconvex and nonlinear MVOP problems. Combinatorial optimization problems involve discrete variables that demand special treatment. Evolutionary algorithms are frequently employed to solve such problems, but they are usually highly related to the problem characteristics and difficult to extend to MVMOPs simply. For example, ant colony algorithms and their variants are customized for TSP problems [13,14], while genetic algorithms require problem-specific encoding and operator design [15].

One common way to tackle MVMOPs is to modify the multi-objective evolutionary algorithm to suit MVMOPs. The critical point of the modification is how to handle variables of different types variables. Some researchers use indirect methods for integer and categorical variables, such as treating them as continuous variables [16], applying relaxation methods, imposing constraint violation penalties [17], and so on. Other researchers adopt direct methods for discrete variables, such as using distributional statistics [18,19], designing specific encoding and recombination operators for different types of variables [20]. However, these algorithms have drawbacks in search efficiency and ability to extract relationships between variables. In this paper, we propose an evolutionary algorithm based on a fully connected weight network model for mixed-variable multi-objective optimization problems, which we call FCWNEA. The main contributions of this paper are as follows.

- A novel distribution model, i.e., a fully connected weight network model (FCWN), is built to characterize the entire decision space of the mixed-variable optimization problem. The distribution of the whole search space is estimated by three attribute values: the activity Act of the network layer reflects the importance of the variables, the number of accesses Acc of the node indicates the search intensity, and the weights W between the nodes capture the correlation among the variables. The attribute values of the model are dynamically adjusted during the evolutionary search.
- An evolutionary algorithm for mixed-variable multi-objective optimization based FCEN model is proposed based on the FCEN model, including the offspring generation mechanism and the distribution model updating method. The whole search process can be viewed as a coevolution process of the solution population and the fully connected weight network. The fully connected

weight network model is updated according to the information feedback obtained from the iterative evolution of the population in the objective space. The offspring population is generated based on the parent population and the fully connected weight network.

- A strategy for dynamic adjustment of node values of continuous variables is designed to achieve a global search for continuous variables. During the search process, continuous variables are discretized. The initial search has a larger granularity and a wider search range. In the evolutionary process, the node values of continuous variables are continuously adjusted to reduce the search granularity and increase the search accuracy.
- Systematic experiments have been conducted to assess the performance of the proposed FCWNEA. The experiments show that the proposed algorithm FCNEA performs well for mixed-variable multi-objective problems, as well as for continuous and mixed-integer multi-objective optimization. Moreover, the algorithm effectively handles complex variable correlations in mixed-variable multi-objective optimization problems.

The rest of the paper is structured as follows. Section 2 reviews the related work and explains the motivation. Section 3 introduces the details of the proposed algorithm. Section 4 presents the test problems used in the experiments. Section 5 conducts the experimental study and discusses the results. Finally, Section 6 summarizes the paper and highlights the main contributions and implications.

2. Literature review

Mixed-variable optimization problems (MVOPs) involve both continuous and discrete variables, which pose challenges for evolutionary algorithms (EAs). In this section, we review the methods for handling mixed variables in EAs, and categorize them into indirect and direct approaches.

Indirect methods: Treating integer variables as continuous and then convert them to integers. The most easily implemented and widely used indirect techniques are rounding and truncation. Jouni Lampinen and Ivan Zelinka [16] proposed a novel optimization method based on the differential evolution algorithm, in which a mixed truncation method is employed to handle integer restrictions and a categorical variable can also be handled as an integer variable by assigning an integer index. Wu et al. [21] employed a novel population initialization technology and dynamic nonlinear scaling to enhance the searchability of DE algorithms. Han and Hu [22] proposed an improved version DE algorithm, in which a local search is utilized to improve the parents. In Ref [23], a general hybrid approach is proposed based on brand-and-bound and non-dominated sorting genetic Algorithm 2 for multi-objective mixed integer nonlinear optimization problems. This approach is straightforward but may not be highly efficient for some problems. Some researchers consider the restrictions on integer and discrete variables as a constraint and the degree of constraint violation as an optimization objective. Liu and Wang [17] transformed a MIP problem into a constrained biobjective optimization problem without integer restrictions. Angira and Babu [24] introduced an additional equality constraint to deal with binary variables. However, this method works better for pseudo-discrete variables and may not be highly efficient for search problems involving categorical variables. Indirect methods are easy to implement and widely used, but they may not be efficient for some problems that involve categorical variables or strict integer constraints.

Direct techniques: Direct techniques directly handle discrete variables without any transformation. Some of these studies are based on widespread EA improvements to codes or operators that allow them to handle mixed-variable problems. For example, Liu et al. [25] designed an improved ACO algorithm to solve MVOPs, named ACO-MV. This method integrates a continuous optimization solver (ACOR), a continuous relaxation approach, and a categorical optimization approach to solve continuous and mixed-variable optimization problems. Dattaa and Figueira [26] proposed a real-integer-categorical-coded method to solve MVOP problems. [27] proposed the MOACOMV algorithm-based ACOMV suitable for solving mixed-variable multi-objective problems. Ref [28] proposed a novel self-adaptive mixed-variable ant colony algorithm for MVOPs. In [29], a real-integer-discrete-coded PSO is proposed, combining binary-coded PSO with real-coded PSO. In [30] proposed an enhanced integer-coded PSO algorithm for feeder reconfiguration. In [31], a mixed-variable encoded PSO algorithm for solving MVOPs is proposed, in which two reproduction methods for continuous variables and discrete variables are proposed. This class of methods is easier to implement, but the search is inefficient. In addition, some studies are based on probability distributions and statistics to optimize categorical variables. In [32], a histogram model-based estimation of the distribution algorithm is developed, in which an adaptive-width histogram model is used to handle continuous variables, and a learning-based histogram model is applied to manage discrete variables. [33] proposed a mixed-variable multi-objective evolutionary algorithm based on the estimation of the distribution algorithm, in which statistics and sampling are used to generate offspring. Jamie et al. [34] utilized Gaussian processes as surrogates in combination with a novel distance metric based on Gower similarity. Yang and Wu [35] proposed a surrogate-based optimization method for mixed-variable aircraft design. However, independent statistics for individual variables do not describe the correlation between variables well, especially for discrete variables. Bayesian methodology is introduced to enhance the optimization efficiency for mixed-variable problems [36,37]. Direct methods directly handle discrete variables without any transformation, but they may require modifying the codes or operators of EAs, or using probability distributions and statistics to optimize categorical variables.

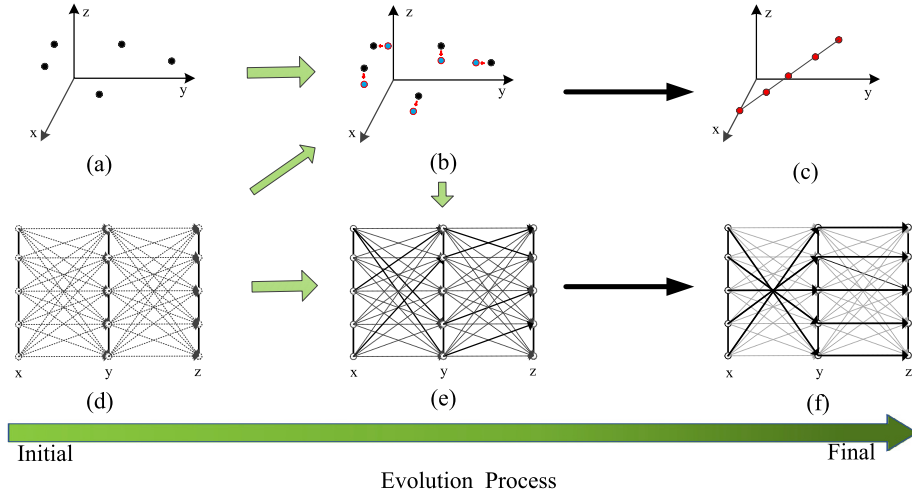


Fig. 1. The basic idea of FCWNEA: The population and FCWN models evolve synergistically and alternately. The FCWN model leverages information from the population search to update its model while concurrently guiding the population to explore promising regions.

3. Proposed algorithm: FCWNEA

In this section, we introduce the fundamental idea and framework of the proposed evolutionary algorithm based on a fully connected weight network model (FCWN) named FCWNEA. Subsequently, we delve into the specifics of its crucial components, encompassing modeling, population structure, reproduction, model updating, and selection.

3.1. Basic idea

Algorithm 1 The framework of FCWNEA.

Require: N (population size), FE_{max} (the maximal number of function evaluations), Pro (the optimization problem)

Ensure: Pop (the final population)

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1: Initialization:  $[FCWN, Pop] \leftarrow \text{Initialization}(N, Pro)$ ; % Initialize the FCWN mode, and initial population  $pop$ .
2: while the number of function evaluations is less than  $FE_{max}$  do
3:   Reproduction:  $Off \leftarrow \text{Reproduction}(Pop, FCWN, FE_{current})$ ; % Generate offspring population  $Off$  according to  $pop$  and  $FCWN$ ;
4:   Update mode:  $FCWN \leftarrow \text{UpdateFCWN}(Pop, Off, FCWN)$ ; % Update the attribute values, and the node value of the layer of the continuous variables.
5:   Environment:  $Pop \leftarrow \text{Selection}(Pop, Off)$ ; % select a new population  $pop$  from  $[Pop, Off]$ , in this paper the Environment selection of NSGA-III is used.
6: end while

```

The core concept of FCWNEA is to leverage a fully connected weight network within an evolutionary algorithm, employing it to statistically the search history and estimate probability distributions. Simultaneously, statistical information within the fully connected network aids in generating offspring solutions within promising regions. The entire search process represents a co-evolutionary interaction between the solution population and the fully connected weight network. As illustrated in Fig. 1, the parent population generates the offspring population based on the fully connected network model. The fully connected network model updates its connection weights by considering the information differences between offspring and parents. In subfigures (d, e, f), the darker connection lines represent higher weights, signifying more potential variable interactions and stronger interdependencies. After iterative optimization, the optimal population and optimal weights of the FCWN model are eventually achieved.

For multi-objective problems, in principle, any environmental selection of the MOEA can be used in the FCWNEA. In this study we use the environment selection of NSGA-III. The framework of FCWNEA is given in Algorithm 1.

3.2. FCWN model and population structure

A fully connected network model with weights, FCWN, is built to estimate the distribution of the decision space. In this part, first, we describe how the fully connected network represents the entire decision space. Then, we discuss how to estimate probability through fully connected networks. Finally, the structure of the population is introduced, and the flow of initialization is presented in Algorithm 2.

3.2.1. How the FCWN represents the decision space

A simple FCWN structure is shown in Fig. 2, in which there are three variables (one continuous variable x_1 , one integer variable x_2 , and one categorical variable x_3).

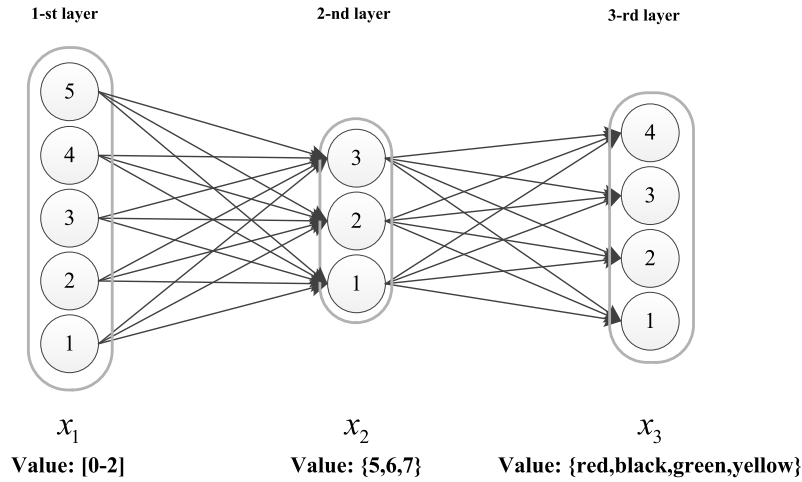


Fig. 2. Illustrating a basic FCWN model with three variable types: continuous variables x_1 , discrete integer variables x_2 , and categorical variables x_3 . The node count per network layer represents the number of values taken by the variables.

- **Variable layer:** Each layer in the network corresponds to a variable. The number of layers in the FCWN is the same as the number of variables D . To illustrate, consider a mixed multi-objective optimization problem with three types of variables: one continuous, one integer, and one categorical. As shown in Fig. 2, the first, second, and third layers correspond to the continuous, integer, and categorical variables, respectively.
- **Value node:** Each layer of variables consists of several nodes. Each node has an index number and a variable value assigned to it. The number of nodes in a layer depends on the nature of the corresponding variable. For integer and categorical variables, the number of possible values is finite, so the number of nodes in the corresponding layer equals the number of values, and each node represents a distinct value. For continuous variables, the value of such variables is infinite. The number of nodes is set to $\text{round}((x^u - x^l)/\delta)$, where x^u and x^l are the upper bound and the lower bound of the variable, respectively, and δ is the interval spacing, for example, in Fig. 2, the range of continuous variables x_1 is [1, 2], and the first layer, corresponding to x_1 , contains 5 nodes, which are assigned [0, 0.5, 1, 1.5, 2] in order. The integer variable x_2 has a set of {5, 6, 7}, and the second layer, corresponding to x_2 , has 3 nodes assigned [5, 6, 7] in turn. The category variable x_3 has a set of {red, black, green, yellow}, and the third layer, corresponding to x_3 , contains 4 nodes assigned [red, black, green, yellow] in order.

Each node has a unique index number and variable value, which serve as the link between the fully connected network and the decision space. For instance, in Fig. 2, a sequence of nodes from the first to the last layer — [3, 2, 4] — corresponds to a solution with variable values [1, 6, yellow]. It is worth noting that continuous variables are discretized in the FCWN model, i.e., a finite number of samples searches continuous space, which contradicts the infinite values of continuous variables. Section 3.4 further explains how the global search for continuous variables can be achieved with a finite number of nodes during the evolution.

3.2.2. How the FCWN model estimates the probability distribution of the decision space

In the preceding section, we established that the FCWN model has the capability to represent the entirety of the decision space. In this section, our objective is to empower the FCWN model to estimate the distribution of the decision space, which requires estimating the search state and identifying promising regions within the decision space. To achieve this, we implement the following process. Within the FCWN framework, we assign three attribute values (weight, access count, and activity) to each of its three components: the connections between nodes, individual nodes, and the layers of variables. Consequently, the FCWN model can effectively estimate the probability distribution of the decision space. Specifically:

- **Weight W :** Each connection between nodes in the adjacent layer is assigned a weight parameter. The larger the weight of the connection is, the more promising the connection, and the stronger the correlation between two variables.
- **Access Count Acc :** The access count of each node tracks how often it has been accessed by previous solutions, revealing the search adequacy and deficiency in different areas.
- **Activity Act :** The activity indicates the probability that the perturbation of this variable improves the quality of the solution.

The FCWN is a statistic of the entire search space and serves as auxiliary information to guide the generation of offspring solutions. The weights between nodes can characterize the correlation between variables. As Fig. 3 shows, the connection line's thickness indicates the connection's weight W . The thicker the line is, the higher the connection probability. The probability distribution curves are shown in Fig. 3 (b) and (c). When the 2-nd node in the i -th layer is selected, the 2-nd node in the $i + 1$ -th layer has a higher probability of being selected. When the 4-th node in layer i is selected, the 4-th node in layer $i + 1$ has a high probability of being selected.

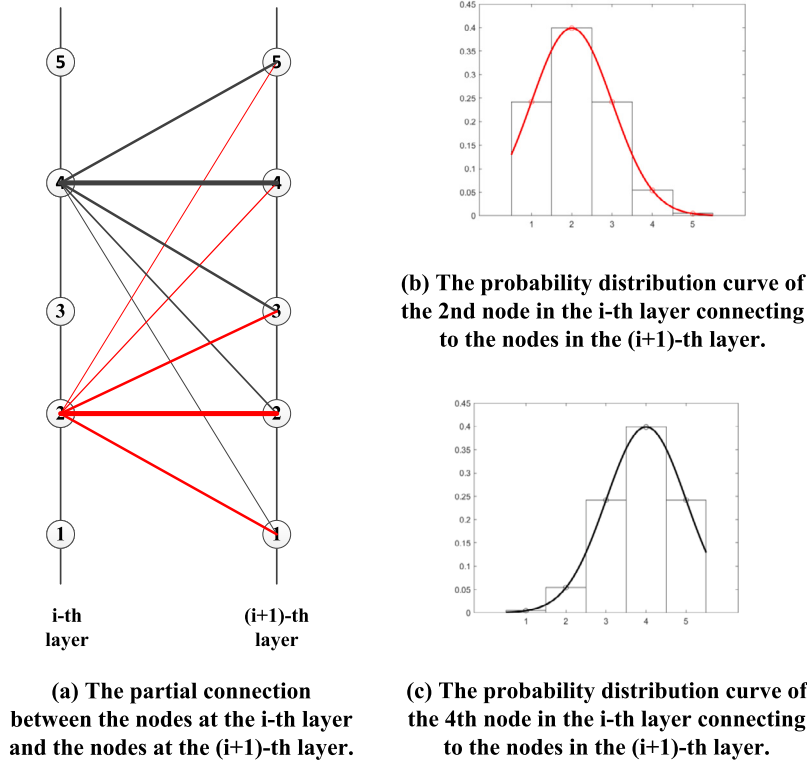


Fig. 3. An example of variable correlation.

	Continuous variables	Integer variables	Categorical variable	Optimization objectives
S_1	$No_1^1, No_1^2, \dots, No_1^{n_1}$	$No_1^{n_1+1}, No_1^{n_1+2}, \dots, No_1^{n_1+n_i}$	$No_1^{n_1+n_i+1}, No_1^{n_1+n_i+2}, \dots, No_1^{n_1+n_i+n_c}$	$f(S_1)^1, f(S_1)^2, \dots$
S_2	$No_2^1, No_2^2, \dots, No_2^{n_1}$	$No_2^{n_1+1}, No_2^{n_1+2}, \dots, No_2^{n_1+n_i}$	$No_2^{n_1+n_i+1}, No_2^{n_1+n_i+2}, \dots, No_2^{n_1+n_i+n_c}$	$f(S_2)^1, f(S_2)^2, \dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
S_k	$No_k^1, No_k^2, \dots, No_k^{n_1}$	$No_k^{n_1+1}, No_k^{n_1+2}, \dots, No_k^{n_1+n_i}$	$No_k^{n_1+n_i+1}, No_k^{n_1+n_i+2}, \dots, No_k^{n_1+n_i+n_c}$	$f(S_k)^1, f(S_k)^2, \dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
S_N	$No_N^1, No_N^2, \dots, No_N^{n_1}$	$No_N^{n_1+1}, No_N^{n_1+2}, \dots, No_N^{n_1+n_i}$	$No_N^{n_1+n_i+1}, No_N^{n_1+n_i+2}, \dots, No_N^{n_1+n_i+n_c}$	$f(S_N)^1, f(S_N)^2, \dots$

Fig. 4. The structure of the population. The population stores the node's index number instead of the variables' values. where S_k denotes the k -th solution, No_k^i denotes the node index of the k -th solution's i -th variable.

Algorithm 2 : $[FCWN, pop] = \text{Initialization}(N, Pro)$.

Require: N (population size), Pro (problem)**Ensure:** P Initialized population pop , their objective Obj and probability model $FCWN$

- 1: **Initialize the structure of FCWN:** Determine the number of layers of $FCWN$ and the nodes in each layer based on the problem.
 - 2: **Initialize the attribute values of FCWN:** The weight W of each connection is set to 1, the access count Acc of each node is set to 0, and the activity Act of each layer is set to 1.
 - 3: **Initialize population randomly:** Randomly generate N solutions. Each solution is a sequence of randomly generated node index numbers of each variable layer in $FCWN$ in turn.
-

3.2.3. Population structure

Combining the characteristics of the FCWN, the population structure was established, as shown in Fig. 4. Unlike the traditional population, the population stores the node's index number instead of the variables' values here. The node index sequence is first transformed into variable values, then the solution can be evaluated. For an D -dimensional ($D = n_r + n_i + n_c$) MVMOP and N solutions, FCWNEA stores the index of the variables and the objective function values of each solution in the population, as shown in Fig. 4, where S_k denotes the k -th solution, No_k^i denotes the node index of the k -th solution's i -th variable. A solution is a node-link from the first to the last layer in the FCWN. The whole population can be regarded as a backbone link network composed of N links.

3.3. Reproduction

As mentioned above, a solution stores an array of index numbers. During reproduction, all operations are based on the index number of the nodes. Three operators are designed to generate offspring solutions, namely, exploitation operator (OP_1), exploration operator (OP_2), and insertion operator (OP_3). Algorithm 3 describes the process of generating the offspring population. The details of the three operators are as follows.

- **Exploitation operator (OP_1):** For a given parent solution $S_k(N o_k^1, \dots, N o_k^i, \dots, N o_K^{(n_r+n_1+n_c)})$, a variable x_i is chosen as a perturbation variable according to the activity Act_i of the variable x_i , which is considered as the fitness. Next, the new node in the i -th layer of FCWN is selected based on the probability Pro_w , as shown in Eq. (2). Afterward, the Act_i of the i -th variable is updated by Eq. (3) to decrease the probability that the variable will be chosen again.

$$Pro_w = (W_{i-1,i}^{No^{j-1},1}, W_{i-1,i}^{No^{j-1},2}, \dots, W_{i-1,i}^{No^{j-1},T^i}) \quad (2)$$

$$Act_i = Act_i / 2; \quad (3)$$

where $W_{i-1,i}^{k,j}$ means the weight of the connection between the k -th node in the $i-1$ layer and the j -th node in the i -th layer of the FCWN. T^i is the total number of the nodes in the i -th layer.

- **Exploration operator (OP_2):** The exploration operator resembles the exploitation operator OP_1 . The difference is that the perturbation variable x_i is selected randomly, and the new node in the i -th layer of the FCWN is selected according to Pro_{Acc} calculated by Eq. (4), which is regarded as the fitness.

$$Pro_{Acc} = (1/Acc_i^1, 1/Acc_i^2, \dots, 1/Acc_i^{T^i}) \quad (4)$$

where Acc_i^j means the access count of the j -th node in the i -th variable, and T^i is the total number of the nodes in the i -th layer.

- **Insertion operator (OP_3):** Randomly select two parent solutions, $S_{k1}(N o_{k1}^1, \dots, N o_{k1}^i, \dots, N o_{k1}^{(n_r+n_1+n_c)})$ and $S_{k2}(N o_{k2}^1, \dots, N o_{k2}^i, \dots, N o_{k2}^{(n_r+n_1+n_c)})$. Two offspring solutions are produced by Eq. (5)

$$\begin{aligned} off_1 &= round(u1 * (S_{k1} - S_{k2})) \\ off_2 &= round(u2 * (S_{k1} - S_{k2})) \end{aligned} \quad (5)$$

where u_1, u_2 are two random numbers in $[0, 1]$.

The ratio of offspring individuals produced by the three operators varied dynamically throughout the optimization process. At the beginning of the optimization, the algorithm has to explore the space globally, and then the exploration operator OP_2 is employed in the largest ratio. With the progressive convergence of the population, it is essential to conduct local exploration, and then the ratio of the operators OP_1 and OP_3 is gradually augmented. Therefore, we devise equations Eq. (6) and Eq. (7) to determine the number of offspring solutions generated by the three operators. In Eq. (6) and Eq. (7), N_{OP_1} , N_{OP_2} and N_{OP_3} are the number of offspring solutions generated by OP_1 , OP_2 and OP_3 respectively. Fig. 5 visually depicts the ratio of the three operators at different phases during the evolutionary process.

$$\begin{cases} N_{OP_1} = \tau * (N - N_{OP_3}) \\ N_{OP_2} = N - (N_{OP_1} + N_{OP_3}) \\ N_{OP_3} = 2 * \lceil \theta * N * \tau / 2 \rceil \end{cases} \quad (6)$$

$$\tau = e^{\frac{FE}{FE_{max}} - 1} \quad (7)$$

Algorithm 3 Reproduction($Pop, FE_{current}$).

Require: Pop (Parent population), FE (The number of current evaluations)

Ensure: Off (Offspring population)

1: $[N_{OP_1}, N_{OP_2}, N_{OP_3}]$ calculated by Eq. (6)

2: **for** $i = 1 \rightarrow N_{OP_1}$ **do**

3: $Off_i \leftarrow$ generated by OP_1 based on P_i

4: **end for**

5: **for** $i = N_{OP_1} + 1 \rightarrow N_{OP_1} + N_{OP_2}$ **do**

6: $Off_i \leftarrow$ generated by OP_2 based on P_i

7: **end for**

8: **for** $i = N_{OP_1} + N_{OP_2} + 1 \rightarrow N$ **do**

9: $[Off_i, Off_{i+1}] \leftarrow$ generated by OP_3 based on P_i and P_{i+1}

10: **end for**

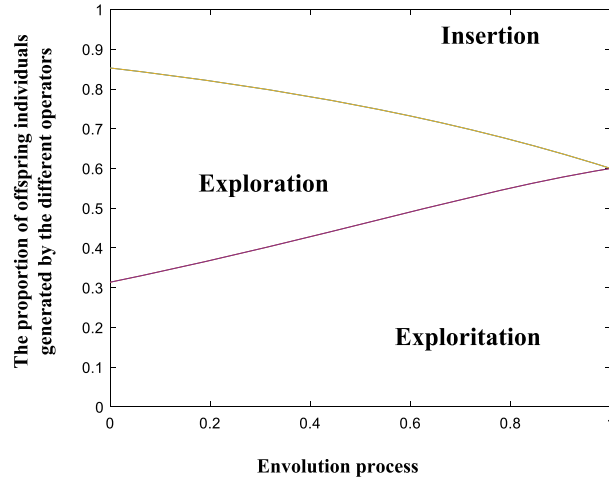


Fig. 5. The proportion of offspring solutions generated by different operators.

3.4. Update FCWN model

The update of the FCWN model involves two primary components: 1) the update of the three attribute values and 2) the update of the node values of the continuous variables.

Algorithm 4 UpdateFCWN(*Pop*, *Off*, *FCWN*).

Require: *Pop* (Parent population), *Off* (offspring population), *FCWN*

Ensure: The updated FCWN

- 1: UpdateAttributes(*Pop*, *Off*, *FCWN*) % Update the three attribute values;
 - 2: UpdateVariableNode(*FCWN*) % Update the node values of the continuous variable layer.
-

3.4.1. The update of the three attribute values of the FCWN

Updating the estimate of the probability distribution of the decision space is actually to update the three attribute values in the FCWN, i.e., the connection weights W , the number of node visits Acc , and the activity of the variable layer Act . The details are described in Algorithm 5. The updated formulas of the three attribute values are described as follows.

- **The update of W :** From the reproduction operation, the one-to-one correspondence between parent and child solutions can be easily established. Therefore, it is easier to find causal relationships between changes in the quality of solutions and changes in variables. If an offspring solution is not inferior to its parent solution, then the associated linkage weights will be updated. The details of the weight update are described in lines 1-9 of Algorithm 5. The weight update formula is shown below.

$$W_{j-1,j}^{No^{j-1},t} = \frac{W_{j-1,j}^{No^{j-1},t'}}{\sum_{t=1}^{T^j} W_{j-1,j}^{No^{j-1},t'}} \quad (8)$$

$$W_{j-1,j}^{No^{j-1},t'} = W_{j-1,j}^{No^{j-1},t} + \Delta_w(t) \quad (9)$$

$$\Delta_w(t) = \frac{1}{\sqrt{2\pi}} e^{\frac{(t-No^j)^2}{2*(K^j/10)}} \quad (10)$$

where $W_{j-1,j}^{No^{j-1},t}$ denotes the weight of the connection between the No^{j-1} -th node in the $j-1$ layer and the t -th node in the j -th layer of the FCWN. $W_{j-1,j}^{No^{j-1},t'}$ is the value of $W_{j-1,j}^{No^{j-1},t}$ before it is normalized. $\Delta_w(t)$ is an incremental change. No^j is the node number of this offspring solution at the j -th variable layer. T^j is the total number of nodes in the j -th layer of the FCWN.

- **The update of Acc :** When a node is visited by a solution once, its Acc is updated by Eq. (11). The update details are described in lines 10-14 of Algorithm 5.

$$Acc_{Off_i}^{No^j} = Acc_{Off_i}^{No^j} + 1 \quad (11)$$

where $Acc_{Off_i}^{No^j}$ is the access count of the selected node of the j -th variable of solution Off_i .

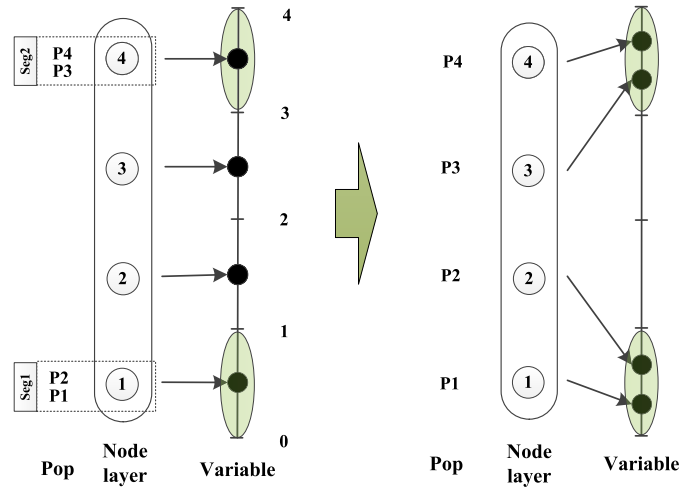


Fig. 6. Update the value of nodes. The values of the nodes are dynamically updated as the population iterates to improve the search accuracy for continuous variables.

- **The update of Act :** The activity of a variable indicates the possibility of improving the solution quality by perturbing this variable in the current state. When generating the offspring population, N_{i_better} and N_{i_worse} are counted, where N_{i_better} is the number of times that the i -th variable perturbation makes the solution quality better and N_{i_worse} is the number of times the i -th variable perturbation makes the solution quality worse. The activity Act_i of the i -th variable is updated by Eq. (12), where D is the number of problem variables.

$$Act_i = \frac{\frac{(1+N_{i_better})^2}{(1+N_{i_worse})}}{\sum_{j=1}^{j=D} \frac{(1+N_{j_better})^2}{1+N_{j_worse}}} \quad (12)$$

Algorithm 5 UpdateAttributes($Pop, Off, FCWN$).

Require: Pop (Parent population), Off (offspring population), $FCWN$

Ensure: $FCWN$ (The updated FCWN model)

```

1: for  $i = 1 \rightarrow N$  do
2:   if  $Off_i < Pop_i$  then
3:     for  $j = 1 \rightarrow D$  do
4:       if  $No_{Off_i}^j = No_{Pop_i}^j$  then
5:         Update the set  $Set\_W$  of link weights by Eq. (8). Where  $Set\_W = \{W_{j-1,j}^{No^{j-1,1}}, \dots, W_{j-1,j}^{No^{j-1,j}}, \dots, W_{j-1,j}^{No^{j-1,T^j}}\}$ ,  $T^j$  is the total number of nodes in  $j$ -th layer.
6:       end if
7:     end for
8:   end if
9: end for
10: for  $i = 1 \rightarrow N$  do
11:   for  $j = 1 \rightarrow D$  do
12:     Update the access count  $AC$  of nodes visited by the offspring population by Eq. (11).
13:   end for
14: end for
15: for  $i = 1 \rightarrow D$  do
16:   Update the variable activity  $Act_i$  of  $i$ -th variable Eq. (12).
17: end for

```

3.4.2. The update of node variable values for continuous variables

For integer and categorical variables, their value is finite, so the number of nodes in a layer is finite. However, the value of continuous variables is infinite, so it is impossible to complete the search of the whole variable range with finite node values. We propose a strategy to dynamically adjust the variable value of node mapping to achieve an efficient search of continuous variables, where the number of nodes in the network layer corresponding to continuous variables is constant. However, the variable value mapped by the same node mapping differs at different stages. As shown in Fig. 6, there are always four nodes, and the variable values corresponding to the nodes gradually change from uniform values at the beginning to partially sparse and partially concentrated values, which also means different search accuracies. The denser the value is, the higher the search accuracy. Algorithm 6 details the continuous variable adjustment strategy.

Algorithm 6 UpdateVariableNode(*Pop*, *FCWN*).**Require:** *FCWN*, *Pop***Ensure:** *FCWN*

```

1: for  $i = 1 \rightarrow N_r$  do
2:   if  $x_i$  is a continuous variable then
3:      $Seg = \{Seg_1, \dots, Seg_J\} \leftarrow$  The nodes of the  $i$ -th variable layer are clustered and partitioned into  $J$  segments based on whether the nodes are visited or not by  $Pop$ ; % In Fig. 6, the nodes are partitioned into 2 segments,  $\{Node_1\}$  and  $\{Node_2\}$ .
4:      $NS = \{NS_1, \dots, NS_J\} \leftarrow$  Counting the number of solutions in each segment of the current population visit; % In Fig. 6,  $NS_1 = 2$ , and  $NS_2 = 2$ .
5:      $k = 0$ ;
6:     for  $j = 1 \rightarrow J$  do
7:        $[L_j, U_j] \leftarrow$  Determine the range of variable values corresponding to segment  $Seg_j$ ; % In Fig. 6, the value range of  $Seg_1$  is  $[0, 1]$ ;
8:        $n = N_{node} * \frac{NS_j}{N}$ ; % The number of realigned nodes in the region  $[L_j, U_j]$ .  $N_{node}$  is the node number of  $i$ -th variable layer. % In Fig. 6, for  $Seg_1$ ,  $n = 4 * 2/4$ ;
9:        $[Node_{k+1}, \dots, Node_{k+n}] \leftarrow$  Take  $n$  values evenly from the region  $[L_j, U_j]$ ; % In Fig. 6, for  $Seg_1$ ,  $[Node_1, Node_2] \leftarrow [0.25, 0.75]$ ;
10:       $k = k + n$ ;
11:     end for
12:   end if
13: end for

```

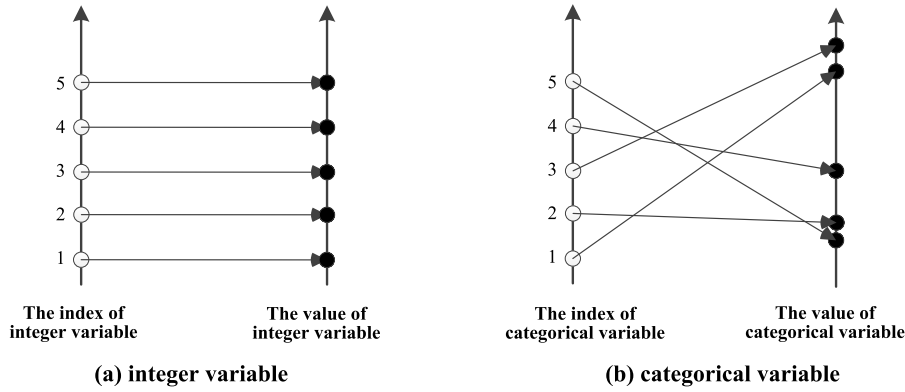


Fig. 7. Integer and categorical variable.

3.5. Computational complexity of one generation of FCWNeA

The most significant difference between FCWNEA and NSGA-III is generating offspring individuals and adding FCWN updates. For NGGAIII, the overall worst-case complexity of the one generation is $O(N^2 \log^{M-2})$ or $O(N^2 M)$ whichever is larger, N is the population size, and M is the number of objectives. Here, we focus on analyzing the complexity of generating offspring individuals and the complexity of FCWN model updates. The 3 operators generate the offspring individuals according to the FCWN model, and the whole generation of the offspring population requires $O(N)$ computations. The update of the FCWN model consists of the update of the attribute values and the update of the node values of the continuous variables. The update of connection weights in lines 1-9 in Algorithm 5 requires $O(2NN_o)$ calculations. The update of the number of visits (line 10-15 in Algorithm 5) takes $O(ND)$ times to compute. The variable activity update requires $N(D)$ computations. Considering that all the above operations are linear in complexity, an additional computational overhead of updating the FCWN model is added relative to the NSGA-III algorithm, but this is acceptable.

4. Artificial mixed-variable multi-objective benchmark function

Most multi-objective optimization benchmark functions are problems with fully continuous variables. In this section, we choose the DTLZ1-7 [38] and UF1-4 [39] problems and convert them to mixed-variable test problems. The number of objectives and variables for the DTLZ problem is scalable. The transformation details are as follows:

The first is to expand the decision space. The problem decision space is stretched by 50 times in the implementation process. (When calculating the objective values, the variable value reduced by 50 is used, which makes the decision space larger but the target space unchanged). Then, the three types of variables are adjusted:

- 1) The continuous variables need not be adjusted;
- 2) The integer variable ranges are all integers between the upper and lower bounds. An index number is assigned to each value of the integer variable, as shown in Fig. 7 (a);
- 3) The values of the categorical variables are the number of categories (1- N), each of which corresponds to a random value behind the number of categories, as shown in Fig. 7 (b).

For integer variables, the search space is discretized, and there still exists a specific neighbor relationship among them. In contrast, categorical variables are employed to discretize the search space further, eliminating the concept of nearest-neighbor relationships. It is worth noting that the influence of categorical variables, when assigned categorical number 1 or categorical number 2, can significantly differ. Categorical variables inherently lack any notion of neighbor relationships.

5. Experimental study

In this section, to examine the performance of FCWNEA, the experiments are divided into five parts.

- To illustrate the effectiveness of the algorithm, it is compared with other state-of-the-art algorithms, including NSGA-III [40], MOEA/D [4], CTAEA [41], RMMEAD [42], MVMO_EDA [33] and PREA [43].
- To demonstrate the adaptability of the FCWNEA algorithm to different types of variables, FCWNEA and NSGA-III were compared in the standard test problem sets DTLZ and UF with different types of variables, including six types of problems: fully continuous, fully integer, fully categorical, continuous+integer, continuous+categorical, and continuous+integer+categorical variable problems.
- Based on the Mv_DTLZ1 problem and the Mv_UF1 problem, the fourth part discusses the ability of FCWNEA to address correlated and uncorrelated variable problems from a microscopic perspective.
- In the fourth part, the effect of the three operators on FCWNEA is discussed.
- Taking the Mv_DTLZ4 problem as an example, the ability of FCWNEA to handle continuous variables is investigated, and investigated the effect of parameter d on FCWNEA.
- Integrated energy system configuration optimization is used as a real case study to test the performance of the proposed FCWNEA.

With respect to experimental settings, the maximum number of evaluation ME is set to 40000 for DTLZ problems and 200000 for UF problems. The population size is set to 100 for 2-objectives problems and 91 for 3-objectives. Simulated binary crossover (SBX) is used by NSGA-II, NSGA-III, MOEA/D, and CTAEA on DTLZ1-7 problems. The differential evolution (DE) mutation operator is used by NSGA-III, MOEA/D, PREA, and CTAEA for UF1-4 problems. The other parameters of the compared algorithms are set according to the original paper of the algorithm.

In the following experimental studies, HV [44] and IGD [45], two widely used performance indicators, are adopted to evaluate the performance of the compared algorithms. All statistical results of the HV metric values are obtained via 20 independent runs. The Wilcoxon rank sum test is also adopted at a significance level of 0.05, where the symbols “+”, “-” and “=” indicate that the result is statistically better, worse, and similar to that obtained by FCWNEA, respectively. All parameters of the compared algorithms are set the same as reported in the original papers, and we use the algorithms implemented in PlatEMO. The relevant code for the proposed algorithm FCWNEA can be obtained at <https://github.com/643260047DNJ/FCWNEA.git>. The experiments are conducted on a computer with the following specifications: Windows 10 operating system and Intel Core i7 processor.

5.1. Comparison with other state-of-the-art algorithms

Table 1 and Table 2 provide the average and the standard deviation of the HV and IGD metric value of the final solutions obtained by PREA, MOEA/D, NSGA-III, RMMEAD, CTAEA, MVMO_EDA and FCWNEA, where the best result on each test instance is shown in a gray background. For all test problems, the number of variables is 12, the 1st to 4th variables are continuous variables, 5th to 8th are integer variables, and 9th to 12th are categorical variables.

In terms of the HV metric, from Table 1, we observe that FCWNEA performs the best on Mv_DTLZ1, 3, 7. PREA performs the best on 3-objective Mv_DTLZ5,6, 5-objective Mv_DTLZ2,6. MOEA/D performs the best on the 2,5-objective Mv_DTLZ5. Regarding the IGD metric, from Table 2, similar to the HV indicator, FCWNEA outperforms the other algorithms. It is inferior to the other algorithms on the Mv_2,4,5 problem. The reason may be that the Mv_DTLZ2,4,5,6 problems easily converge. The Mv_DTLZ2,4 problem itself easily converges, and even if there are integer variables and classification variables, the original algorithm obtains good optimization results through the integer operation. For Mv_DTLZ5,6, the parent front is a one-dimensional fluid even if the optimization objective number is higher than 2 [38]. However, for the Mv_UF Problem, FCWNEA performs the best on all problems in both the HV and IGD metrics. The Mv_UF problem involves highly correlated variables. The strong correlations among the categorical variables make the search difficult for other algorithms. The FCWN model can better represent the correlations among the categorical variables and achieve a higher search efficiency.

In general, the proposed algorithm performs well for mixed-variable multi-objective problems. However, the other algorithms do not achieve satisfactory results for MVMOP problems by simple rounding operations on the SBX and DE operators. In addition, it is worth noting that FCWNEA is robust to both variable uncorrelated problems (Mv_DTLZ problems) and variable correlated problems (Mv_UF problems). Usually, researchers choose the SBX operator for variable uncorrelated problems and DE for variable correlated problems, which seem to be inappropriate.

5.2. Comparison with NSGA-III on different types of problems

To demonstrate the adaptability of the algorithm to problems with different types of variables, the FCWNEA and NSGA-III are compared to the DTLZ and UF problems. The DTLZ and UF problems were modified into six types of problems, namely, fully

Table 1

Average and standard deviation of the HV metric values obtained by the compared algorithms on the Mv_DTLZ 1-7 and Mv_Uf 1-4 problems. The value in bold on the gray background is the best result among the compared algorithms based on the mean metric value. '+', '-' and '=' indicate that the result is significantly better, significantly worse and statistically similar to that of FCWNEA, respectively.

Problem	M	PREA	MOEA/D	NSGA-III	RMEDA	CTAEA	MVMO_EDA	FCWNEA
Mv_DTLZ1	2	2.3052e-1 (2.98e-1) -	1.7254e-1 (2.78e-1) -	1.7304e-1 (2.79e-1) -	0.0000e+0 (0.00e+0) -	2.3477e-1 (2.95e-1) =	0.0000e+0 (0.00e+0) -	5.7562e-1 (1.89e-3)
	3	3.5179e-1 (4.21e-1) =	1.6765e-1 (3.53e-1) -	3.3536e-1 (4.33e-1) =	0.0000e+0 (0.00e+0) -	2.6661e-1 (3.96e-1) -	0.0000e+0 (0.00e+0) -	8.2920e-1 (1.26e-2)
	5	6.6535e-1 (4.16e-1) -	1.4679e-1 (3.22e-1) -	1.0479e-1 (1.90e-1) -	0.0000e+0 (0.00e+0) -	1.6400e-1 (3.13e-1) -	0.0000e+0 (0.00e+0) -	9.6680e-1 (5.89e-3)
Mv_DTLZ2	2	3.4514e-1 (8.07e-4) -	3.4591e-1 (9.21e-4) =	3.4581e-1 (1.32e-3) =	3.1333e-1 (1.16e-2) -	3.4604e-1 (6.09e-4) =	9.4291e-2 (1.97e-2) -	3.4665e-1 (2.84e-4)
	3	5.5704e-1 (2.52e-3) =	5.5765e-1 (1.28e-3) -	5.5909e-1 (5.96e-4) =	4.2306e-1 (1.32e-2) -	5.4452e-1 (3.10e-3) -	7.2642e-2 (2.32e-2) -	5.5912e-1 (1.78e-4)
	5	7.7708e-1 (2.72e-3) =	7.7330e-1 (7.26e-4) -	7.7366e-1 (9.01e-4) -	9.6371e-2 (2.08e-2) -	7.3855e-1 (5.64e-3) -	8.8194e-2 (6.32e-2) -	7.7460e-1 (3.08e-4)
Mv_DTLZ3	2	6.5206e-2 (6.89e-2) -	6.2198e-2 (1.32e-1) -	1.5335e-1 (1.39e-1) -	0.0000e+0 (0.00e+0) -	1.7859e-1 (1.22e-1) -	0.0000e+0 (0.00e+0) -	3.4554e-1 (1.31e-3)
	3	1.7485e-1 (1.57e-1) -	5.6203e-2 (1.77e-1) -	2.9371e-1 (1.87e-1) -	0.0000e+0 (0.00e+0) -	2.7343e-1 (1.87e-1) -	0.0000e+0 (0.00e+0) -	5.5715e-1 (9.54e-4)
	5	6.0553e-1 (1.84e-1) =	8.1024e-2 (2.43e-1) -	3.9552e-1 (1.60e-1) -	0.0000e+0 (0.00e+0) -	2.2848e-1 (2.29e-1) -	0.0000e+0 (0.00e+0) -	7.5740e-1 (2.66e-2)
Mv_DTLZ4	2	2.6781e-1 (1.23e-1) -	2.8529e-1 (1.06e-1) =	3.2092e-1 (8.08e-2) =	2.5763e-1 (2.38e-2) -	3.4547e-1 (1.59e-3) =	1.4581e-2 (2.10e-2) -	3.4575e-1 (1.75e-3)
	3	5.3734e-1 (6.77e-2) =	3.7369e-1 (2.15e-1) =	4.2348e-1 (1.60e-1) =	3.5733e-1 (2.58e-2) -	5.4315e-1 (3.19e-3) -	1.1573e-2 (2.55e-2) -	5.5765e-1 (9.68e-4)
	5	6.4374e-1 (1.38e-1) =	6.7960e-1 (8.07e-2) =	6.9739e-1 (6.13e-2) -	2.5507e-1 (4.50e-2) -	7.2536e-1 (7.51e-3) -	1.7863e-2 (2.44e-2) -	7.5773e-1 (1.98e-3)
Mv_DTLZ5	2	3.4660e-1 (6.74e-4) =	3.4692e-1 (3.20e-4) =	3.4651e-1 (4.95e-4) =	3.0760e-1 (1.23e-2) -	3.4657e-1 (7.66e-4) =	1.2514e-1 (2.79e-2) -	3.4638e-1 (1.29e-3)
	3	1.9961e-1 (2.86e-4) +	1.8138e-1 (4.09e-4) -	1.9193e-1 (1.65e-3) -	1.5728e-1 (9.11e-3) -	1.9476e-1 (1.59e-3) -	2.8660e-2 (1.66e-2) -	1.9642e-1 (2.56e-3)
	5	1.0525e-1 (3.11e-3) +	1.2371e-1 (6.17e-4) +	1.0283e-1 (4.24e-3) +	4.2921e-2 (1.93e-2) -	1.1209e-1 (3.17e-3) +	2.9415e-2 (3.11e-2) -	9.8552e-2 (3.80e-3)
Mv_DTLZ6	2	3.1258e-1 (1.10e-1) -	1.7326e-1 (1.83e-1) =	3.4722e-1 (8.05e-8) +	0.0000e+0 (0.00e+0) -	3.1235e-1 (1.10e-1) -	0.0000e+0 (0.00e+0) -	3.4625e-1 (1.19e-3)
	3	1.9982e-1 (8.88e-5) +	7.2742e-2 (9.39e-2) -	1.9114e-1 (1.89e-3) -	0.0000e+0 (0.00e+0) -	1.7853e-1 (3.84e-3) -	0.0000e+0 (0.00e+0) -	1.9774e-1 (5.55e-4)
	5	9.1299e-2 (3.28e-4) =	7.4588e-2 (6.42e-2) =	6.4799e-2 (4.48e-2) =	0.0000e+0 (0.00e+0) -	5.6699e-2 (4.89e-2) =	0.0000e+0 (0.00e+0) -	7.9073e-2 (4.17e-2)
Mv_DTLZ7	2	1.8623e-1 (3.94e-2) -	1.8456e-1 (5.73e-2) -	2.4097e-1 (5.57e-3) =	8.9830e-2 (4.78e-2) -	2.3789e-1 (7.47e-3) -	3.9841e-2 (1.54e-2) -	2.4271e-1 (9.45e-5)
	3	2.5114e-1 (2.61e-2) =	2.2875e-1 (2.88e-2) -	2.6328e-1 (1.24e-2) -	5.1016e-2 (3.86e-2) -	2.6254e-1 (9.76e-3) -	1.2364e-2 (2.10e-3) -	2.7448e-1 (1.53e-3)
	5	2.4170e-1 (1.78e-2) =	2.7072e-2 (3.56e-2) -	2.3368e-1 (1.01e-2) =	4.7250e-3 (7.37e-3) -	2.2852e-1 (1.78e-2) =	4.2751e-4 (2.27e-4) -	2.3764e-1 (5.54e-3)
Mv_UF1	2	6.2656e-1 (1.20e-2) -	4.7636e-1 (7.08e-2) -	2.9592e-1 (1.28e-1) -	6.2646e-1 (1.34e-2) -	6.4480e-1 (2.08e-2) -	5.1666e-1 (3.62e-2) -	6.8601e-1 (1.49e-2)
Mv_UF2	2	6.9440e-1 (6.30e-3) -	6.6582e-1 (1.90e-2) -	3.8761e-1 (7.36e-2) -	6.6572e-1 (7.23e-3) -	6.9654e-1 (4.55e-3) -	5.6884e-1 (2.37e-2) -	7.0911e-1 (1.87e-3)
Mv_UF3	2	3.8124e-1 (2.79e-2) -	3.4840e-1 (3.52e-2) -	1.1790e-2 (2.47e-2) -	7.4438e-2 (2.19e-2) -	4.3070e-1 (4.28e-2) -	2.5275e-1 (8.49e-2) -	4.6331e-1 (4.31e-2)
Mv_UF4	2	4.0252e-1 (1.55e-3) -	3.6644e-1 (6.75e-3) -	3.3758e-1 (9.40e-3) -	3.5635e-1 (6.10e-3) -	3.9249e-1 (1.29e-3) -	2.3224e-1 (7.92e-3) -	4.0964e-1 (1.23e-3)

continuous variable, fully integer variable, fully categorical variable, continuous+integer variable, continuous+categorical variable, and continuous+integer+categorical problems. These six types of problems are represented by six prefixes: R_, I_, C_, R+I_, R+C_, and Mv_. For all problems, the number of variables is set to 12. For each problem, the number of different types of variables included in the problem is the same. Table 3 presents the average and standard deviation of HV metric values of the final solution obtained by FCWNEA and NSGA-III. From Table 3, the following observations can be made.

Table 2

Average and standard deviation of the IGD metric values obtained by the compared algorithms on the Mv_DTLZ 1-7 and Mv_Uf 1-4 problems. The value in bold on the gray background is the best result among the compared algorithms based on the mean metric value. '+', '-' and '=' indicate that the result is significantly better, significantly worse and statistically similar to that of FCWNEA, respectively.

Problem	M	PREA	MOEA/D	NSGA-III	RMEDA	CTAEA	MVMO_EDA	FCWNEA
Mv_DTLZ1	2	1.7616e+0 (1.74e+0) -	2.2488e+0 (2.00e+0) -	1.1119e+0 (1.28e+0) -	6.5418e+1 (1.21e+1) -	5.0365e-1 (7.27e-1) =	3.5686e+1 (1.42e+1) -	3.5589e-3 (6.32e-4)
	3	7.6006e-1 (9.72e-1) =	2.9416e+0 (2.10e+0) -	4.5395e-1 (3.89e-1) =	4.4979e+1 (6.95e+0) -	5.1433e-1 (3.91e-1) -	4.6703e+1 (1.68e+1) -	2.5935e-2 (6.63e-3)
	5	2.4283e-1 (2.64e-1) -	2.1704e+0 (2.27e+0) -	5.6293e-1 (1.84e-1) -	3.3481e+1 (5.78e+0) -	6.6277e-1 (5.05e-1) -	1.2829e+1 (5.92e+0) -	6.9951e-2 (4.90e-3)
Mv_DTLZ2	2	5.3546e-3 (3.13e-4) -	4.4567e-3 (3.02e-4) =	4.5036e-3 (4.44e-4) =	2.9481e-2 (9.95e-3) -	4.5538e-3 (9.44e-5) =	2.0340e-1 (1.88e-2) -	4.6193e-3 (3.20e-4)
	3	5.9434e-2 (1.81e-3) -	5.4544e-2 (6.45e-5) +	5.4504e-2 (5.90e-5) +	1.1695e-1 (5.11e-3) -	6.4383e-2 (1.74e-3) -	3.3655e-1 (2.52e-2) -	5.4737e-2 (9.68e-5)
	5	2.2155e-1 (3.49e-3) -	2.1230e-1 (2.13e-4) =	2.1241e-1 (1.68e-4) -	5.5059e-1 (2.87e-2) -	2.4429e-1 (3.05e-3) -	6.2354e-1 (1.47e-1) -	2.1216e-1 (2.57e-4)
Mv_DTLZ3	2	4.8066e-1 (3.51e-1) -	8.0532e-1 (6.83e-1) -	3.0934e-1 (3.70e-1) =	1.6782e+1 (1.15e+0) -	1.5492e-1 (1.17e-1) =	6.1883e+0 (2.64e+0) -	5.4208e-3 (1.73e-3)
	3	4.4587e-1 (3.41e-1) -	1.0533e+0 (7.55e-1) -	2.5149e-1 (2.22e-1) -	1.3743e+1 (1.33e+0) -	2.5936e-1 (2.20e-1) -	1.9615e+1 (4.50e+0) -	5.5809e-2 (1.10e-3)
	5	2.7627e-1 (6.95e-2) -	1.2193e+0 (7.86e-1) -	3.4868e-1 (6.28e-2) -	9.6949e+0 (2.22e+0) -	6.0450e-1 (3.29e-1) -	7.7666e+0 (2.27e+0) -	2.1702e-1 (6.01e-3)
Mv_DTLZ4	2	2.2664e-1 (3.56e-1) =	1.6356e-1 (3.07e-1) =	7.7882e-2 (2.33e-1) -	8.5649e-2 (2.07e-2) -	4.8319e-3 (7.71e-4) +	6.7192e-1 (1.63e-1) -	7.0319e-3 (3.46e-3)
	3	1.0519e-1 (1.53e-1) =	4.1966e-1 (4.12e-1) =	3.3866e-1 (3.23e-1) =	2.2316e-1 (2.24e-2) -	6.8079e-2 (2.23e-3) -	6.8506e-1 (1.37e-1) -	5.6445e-2 (1.00e-3)
	5	4.6900e-1 (2.17e-1) -	3.9470e-1 (1.39e-1) =	3.5260e-1 (1.08e-1) =	4.9504e-1 (1.85e-2) -	2.7209e-1 (5.05e-3) -	7.7521e-1 (7.63e-2) -	2.5123e-1 (1.44e-3)
Mv_DTLZ5	2	4.7471e-3 (1.89e-4) =	4.0346e-3 (7.84e-5) +	4.0626e-3 (8.36e-5) +	3.3514e-2 (1.02e-2) -	4.1379e-3 (1.81e-4) +	1.7953e-1 (2.70e-2) -	5.3625e-3 (1.72e-3)
	3	4.7861e-3 (1.52e-4) +	3.3806e-2 (9.95e-5) -	1.5178e-2 (8.85e-4) -	4.6429e-2 (7.05e-3) -	1.2349e-2 (2.66e-3) -	2.1940e-1 (3.36e-2) -	9.7286e-3 (3.62e-3)
	5	9.9914e-2 (1.30e-2) +	2.6156e-2 (5.54e-4) +	1.1236e-1 (1.45e-2) +	2.3468e-1 (4.35e-2) -	1.0528e-1 (1.79e-2) +	2.8462e-1 (2.08e-1) -	1.5581e-1 (3.97e-2)
Mv_DTLZ6	2	7.1833e-2 (2.12e-1) =	3.4534e-1 (3.59e-1) =	3.9662e-3 (3.13e-8) +	3.0287e+0 (4.58e-1) -	7.1275e-2 (2.13e-1) -	6.3092e+0 (1.02e+0) -	5.4455e-3 (1.66e-3)
	3	4.6670e-3 (9.95e-5) +	4.3238e-1 (3.43e-1) -	1.9188e-2 (2.22e-3) -	3.4604e+0 (5.64e-1) -	4.0641e-2 (6.30e-3) -	7.4214e+0 (4.71e-1) -	7.6997e-3 (1.03e-3)
	5	2.2802e-1 (4.25e-2) =	3.1385e-1 (3.72e-1) =	3.9033e-1 (2.82e-1) =	3.5635e+0 (4.48e-1) -	4.6661e-1 (3.34e-1) =	6.1100e+0 (5.14e-1) -	2.9742e-1 (1.34e-1)
Mv_DTLZ7	2	3.3933e-1 (2.31e-1) =	2.6544e-1 (3.03e-1) -	8.1254e-3 (9.17e-3) =	4.4409e-1 (1.88e-1) -	1.3414e-2 (1.19e-2) -	5.9056e-1 (1.23e-1) -	5.4970e-3 (3.96e-4)
	3	2.5391e-1 (2.44e-1) =	1.6612e-1 (3.07e-2) -	1.1194e-1 (9.14e-2) -	7.8883e-1 (3.07e-1) -	8.3303e-2 (1.12e-2) -	7.6185e-1 (8.30e-2) -	7.2468e-2 (3.66e-3)
	5	3.4257e-1 (9.15e-2) =	9.5305e-1 (2.11e-1) -	3.8148e-1 (2.10e-2) -	1.5311e+0 (5.13e-1) -	4.3753e-1 (2.37e-1) =	1.1464e+0 (3.40e-1) -	3.7801e-1 (1.98e-2)
Mv_UF1	2	7.9664e-2 (9.81e-3) -	2.2682e-1 (7.84e-2) -	3.5779e-1 (1.55e-1) -	6.7591e-2 (9.48e-3) -	2.4163e+0 (6.43e-1) 1	1.4522e-1 (2.81e-2) -	3.3197e-2 (1.61e-2)
Mv_UF2	2	3.3002e-2 (1.10e-2) -	7.3341e-2 (4.00e-2) -	2.4770e-1 (5.88e-2) -	4.4153e-2 (4.90e-3) -	3.0715e+0 (5.39e-1) =	1.3144e-1 (3.84e-2) -	1.5268e-2 (2.30e-3)
Mv_UF3	2	3.2811e-1 (3.96e-2) -	3.4864e-1 (4.27e-2) -	9.2788e-1 (1.17e-1) -	6.5015e-1 (3.28e-2) -	3.6186e+0 (4.89e-1) -	4.4664e-1 (1.09e-1) -	2.2943e-1 (6.78e-2)
Mv_UF4	2	3.3061e-2 (1.42e-3) -	5.3400e-2 (4.72e-3) -	7.5926e-2 (6.65e-3) -	6.4233e-2 (4.40e-3) -	3.3642e+0 (3.23e-1) -	1.6344e-1 (2.13e-2) -	2.7365e-2 (1.53e-3)

First, regarding the three types of problems, namely, fully categorical variables, continuous + categorical variables, and continuous + integer + categorical variables, FCWNEA outperforms NSGA-III on almost all problems, except problem R_DTLZ2. It can be observed that all of these problems include categorical variables. That is because categorical variables break the nearest-neighbor relationship of the values taken by the variables, which leads to the degradation of the search capability of the SBX, PM, and DE oper-

Table 3

Average and standard deviation of the HV metric values obtained by FCWNEA and NSGA-III problems on six groups problems. These six sets of problems are six versions of DTLZ and UF problems that have different types of variables. Specifically, the prefixes R, I, C, R+I, R+C, and Mv denote continuous, integer, categorical, continuous+integer, continuous+integer+categorical, and continuous+integer+categorical variables, respectively. The value in bold on the gray background is the best result among the compared algorithms based on the mean metric value. '+', '-', and '=' indicate that the result is significantly better, significantly worse and statistically similar to that of FCWNEA, respectively.

a			b			c		
Problem	FCWNEA	NSGA-III	Problem	FCWNEA	NSGA-III	Problem	FCWNEA	NSGA-III
R_DTLZ1	8.1148e-1 (6.70e-2)	7.8594e-1 - (6.67e-2)	I_DTLZ1	7.0745e-1 (2.94e-1)	8.4151e-1 + (0.00e+0)	C_DTLZ1	7.1870e-1 (1.29e-1)	0.0000e+0 - (0.00e+0)
R_DTLZ2	5.5920e-1 (3.26e-4)	5.5958e-1 + (1.22e-5)	I_DTLZ2	5.5880e-1 (1.56e-4)	5.5922e-1 + (0.00e+0)	C_DTLZ2	5.4776e-1 (1.29e-3)	5.5272e-1 + (1.88e-3)
R_DTLZ3	5.5743e-1 (2.14e-3)	5.3347e-1 - (1.23e-2)	I_DTLZ3	5.2224e-1 (6.48e-2)	5.5922e-1 + (5.47e-6)	C_DTLZ3	4.5529e-1 (9.57e-2)	3.7793e-2 - (7.95e-2)
R_DTLZ4	5.5580e-1 (1.40e-3)	5.1562e-1 - (9.82e-2)	I_DTLZ4	2.8676e-1 (4.68e-6)	2.4712e-1 - (5.42e-2)	C_DTLZ4	1.2270e-1 (1.17e-5)	9.6307e-2 - (7.93e-3)
R_DTLZ5	1.9424e-1 (1.13e-3)	1.9451e-1 = (9.92e-4)	I_DTLZ5	1.9640e-1 (4.32e-4)	1.9256e-1 - (1.25e-3)	C_DTLZ5	1.8561e-1 (3.73e-3)	1.8366e-1 - (2.27e-3)
R_DTLZ6	1.9579e-1 (5.24e-4)	1.9205e-1 - (2.15e-3)	I_DTLZ6	1.9633e-1 (4.99e-4)	1.9140e-1 - (1.42e-3)	C_DTLZ6	1.9104e-1 (3.72e-4)	1.8527e-1 - (1.29e-3)
R_DTLZ7	2.7218e-1 (2.73e-3)	2.6389e-1 - (1.47e-2)	I_DTLZ7	2.7478e-1 (2.50e-3)	6.6538e-2 - (9.45e-2)	C_DTLZ7	2.5842e-1 (9.82e-3)	1.9102e-2 - (2.72e-2)
R_UF1	7.1485e-1 (4.42e-4)	7.1543e-1 + (6.74e-4)	I_UF1	7.1486e-1 (6.57e-4)	7.1003e-1 - (8.95e-4)	C_UF1	6.9359e-1 (1.10e-2)	6.3261e-1 - (1.66e-2)
R_UF2	7.1227e-1 (1.39e-3)	7.1123e-1 - (6.57e-4)	I_UF2	7.1303e-1 (1.07e-3)	7.0673e-1 - (1.10e-3)	C_UF2	7.0481e-1 (1.80e-3)	6.6238e-1 - (5.72e-3)
R_UF3	5.8919e-1 (2.18e-2)	4.3122e-1 - (1.40e-1)	I_UF3	5.7449e-1 (5.53e-2)	2.1201e-1 - (6.74e-2)	C_UF3	3.9755e-1 (5.41e-2)	3.1309e-1 - (3.68e-2)
R_UF4	4.0952e-1 (1.06e-3)	3.8264e-1 - (5.23e-3)	I_UF4	4.0782e-1 (8.28e-4)	3.8262e-1 - (4.60e-3)	C_UF4	4.0178e-1 (1.53e-3)	3.5616e-1 - (3.64e-3)

d			e			f		
Problem	FCWNEA	NSGA-III	Problem	FCWNEA	NSGA-III	Problem	FCWNEA	NSGA-III
R+I_DTLZ1	7.6700e-1 (6.29e-2)	8.3980e-1 + (3.42e-3)	R+C_DTLZ1	7.8760e-1 (6.19e-2)	0.000e+0 - (0.00e+0)	Mv_DTLZ1	7.9422e-1 (4.88e-2)	2.4076e-1 - (3.46e-1)
R+I_DTLZ2	5.5930e-1 (1.35e-4)	5.5961e-1 + (1.07e-5)	R+C_DTLZ2	3.4455e-1 (6.23e-4)	NaN - (NaN)	Mv_DTLZ2	5.5797e-1 (4.56e-4)	5.5897e-1 + (6.36e-4)
R+I_DTLZ3	5.5066e-1 (1.69e-3)	5.5731e-1 + (4.30e-3)	R+C_DTLZ3	5.3400e-1 (3.50e-2)	NaN - (NaN)	Mv_DTLZ3	5.4013e-1 (8.36e-3)	2.9895e-1 - (2.52e-1)
R+I_DTLZ4	5.5176e-1 (8.39e-3)	5.1506e-1 - (9.92e-2)	R+C_DTLZ4	5.4912e-1 (7.64e-3)	1.7884e-4 - (3.10e-4)	Mv_DTLZ4	5.4961e-1 (6.06e-3)	5.5942e-1 + (8.97e-5)
R+I_DTLZ5	1.9586e-1 (6.80e-4)	1.9482e-1 = (1.11e-3)	R+C_DTLZ5	1.9200e-1 (2.55e-3)	0.000e+0 - (0.00e+0)	Mv_DTLZ5	1.9451e-1 (1.35e-3)	1.9319e-1 = (1.25e-3)
R+I_DTLZ6	1.9643e-1 (5.11e-4)	1.9128e-1 - (1.34e-3)	R+C_DTLZ6	1.9608e-1 (5.61e-4)	0.000e+0 - (0.00e+0)	Mv_DTLZ6	1.9526e-1 (9.16e-4)	1.9117e-1 - (1.30e-3)
R+I_DTLZ7	2.7182e-1 (3.21e-3)	1.2432e-1 - (9.36e-2)	R+C_DTLZ7	2.7068e-1 (4.68e-3)	0.000e+0 - (0.00e+0)	Mv_DTLZ7	2.7144e-1 (2.16e-3)	6.6243e-2 - (5.88e-2)
R+I_UF1	7.1196e-1 (8.78e-3)	7.1324e-1 + (1.52e-3)	R+C_UF1	6.9682e-1 (9.63e-3)	5.8253e-1 - (8.62e-2)	Mv_UF1	6.9884e-1 (3.76e-3)	6.5323e-1 - (2.22e-2)
R+I_UF2	7.1067e-1 (1.75e-3)	7.0956e-1 - (1.07e-3)	R+C_UF2	7.0827e-1 (1.12e-3)	6.5769e-1 - (6.45e-3)	Mv_UF2	7.0587e-1 (1.58e-3)	6.8551e-1 - (4.82e-3)
R+I_UF3	5.7719e-1 (2.14e-2)	2.9696e-1 - (1.13e-1)	R+C_UF3	4.0572e-1 (1.31e-3)	3.5319e-1 - (8.11e-3)	Mv_UF3	4.4560e-1 (3.60e-2)	4.2700e-1 - (2.02e-2)
R+I_UF4	4.0973e-1 (6.95e-4)	3.8244e-1 - (6.88e-3)	R+C_UF4	4.9636e-1 (2.98e-2)	4.1502e-1 - (5.29e-2)	Mv_UF4	4.0638e-1 (7.07e-4)	3.7652e-1 - (1.17e-3)

ators. However, FCWNEA uses fully connected weight networks to estimate the distribution of the decision space and the knowledge accumulated during the search process to generate offspring solutions, which is very beneficial for MVMOPs.

Second, FCWNEA and NSGA-III are comparable with regard to the fully continuous, fully integer, and continuous+integer problems. NSGA-II outperforms FCWNEA on I₁ and R+I₁ DTLZ1-3 and R_DTLZ 2,5. FCWNEA outperforms NSGA-III on I₁ and R+I₁ DTLZ4-7 and R_DTLZ1,3-4,6-7. That indicates that FCWNEA shows generally better performance than NSGA-III on problems with categorical variables and is also competitive on continuous problems.

Third, for UF problems, FCWNEA is superior to NSGA-III in almost all problems except for R_UF1 and R+I_UF1. That means that FCWNEA can efficiently optimize the variable correlation problem, regardless of whether the variable is a continuous, integer variable, or categorical variable.

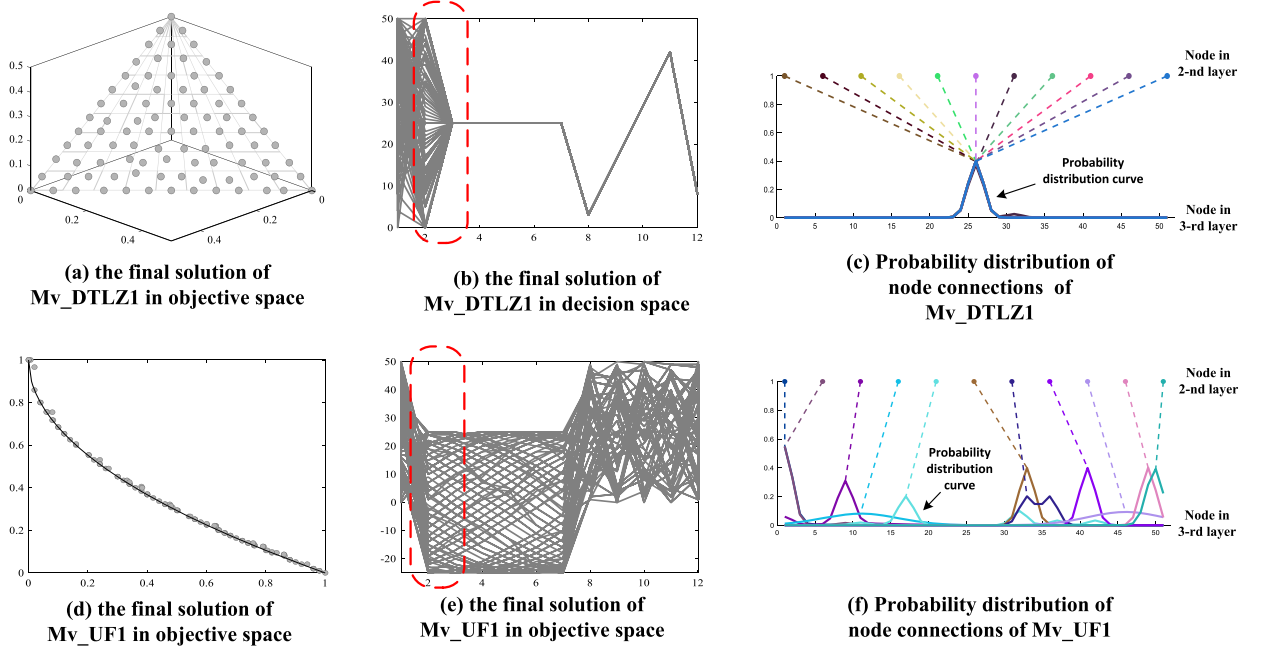


Fig. 8. The experimental results optimized by FCWNEA on Mv_DTLZ1 and Mv_UF1.

5.3. The ability to handle problems with correlated and uncorrelated variables

In this part, we investigate the effect of the FCWN model on the problem of correlated and uncorrelated variables from a microscopic perspective, that is, the probability distribution of a layer of network nodes. The FCWNEA algorithm is further investigated in depth on problems Mv_DTLZ1 and Mv_UF1. For Mv_DTLZ1, its variables are divided into convergence-related and diversity-related variables, and there is no interrelationship between the variables. Mv_UF1 demonstrates the variable correlation problem. Both problems have 12 variables, amongst which the 1st to 4th are continuous variables, the 5th to 8th are integer variables, and the 9th to 12th are categorical variables. For Mv_DTLZ1, the 3rd to 12th are also convergence-related variables.

Fig. 8 presents the final results obtained by FCWNEA on problems Mv_DTLZ1 and Mv_UF1. In Fig. 8, (a) and (d) are the results of the objective values, and (b) and (e) are the results of the decision variables. Fig. 8 (c) and (f) are the probability distributions between some nodes in the second variable layer and all nodes in the third variable layer.

As seen from Fig. 8 (b), for the Mv_DTLZ1 problem, when the population converges to the optimal front, the values of all convergence-related variables (3rd to 12th variables) are concentrated on one point. Here, we focus on the relationship between the values of the second variable and the third variable; the third variable is decentralized, and the fourth variable is centralized. Fig. 8 (c) shows the probability distribution of the connection between the layer 2 nodes and layer 3 nodes in the FCWN model. It can be observed that the probability distribution of any node in layer 2 connecting to the nodes in layer 3 is almost the same, and the 26th node in layer 3 has the highest probability of being selected.

However, for the Mv_UF1 problem, the first variable is a diversity-related variable, and the 2nd to 12th variables are convergence-related variables, but there are correlations between the diversity-related variables and the convergence-related variables. As shown in Fig. 8 (e), unlike Mv_DTLZ1's convergence-related variables, which are clustered in one value domain, all variables of Mv_UF1 are widely distributed throughout the value domain. Continuous variables and categorical variables have nearest neighbor relationships. Therefore, the connection between the 1st to the 8th variable has a strong regularity. For the 8-12 variables, because the categorical variables do not have a close neighbor relationship, the connection looks disordered. However, there is still an inherent correlation. Fig. 8 (f) shows the probability distribution of the connection between layer 2 nodes and layer 3 nodes in the FCWN model. The probability distribution of each node in layer 2 to the nodes in layer 3 is different. Fig. 8 (a) and (b) show the optimization results for Mv_DTLZ1 and Mv_UF1, and it can be seen that both converge to parent front (PF) with uniform coverage. Overall, the FCWN model of FCWNEA is effective for both variable uncorrelated and variable correlated mixed-variable multi-objective problems.

5.4. The effect of three generated operators

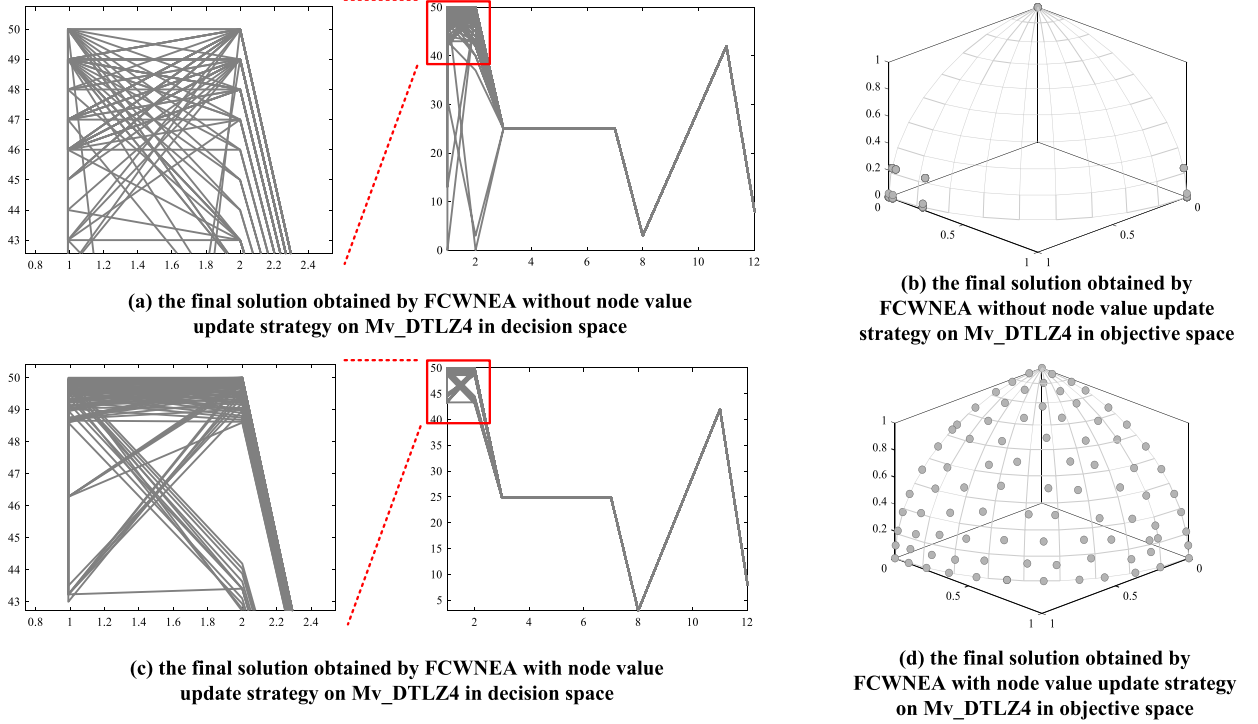
In section 3.4, three operators are designed to generate offspring solutions. In this part, we focus on the effects of different operators used individually and in combination. Five versions of FCWNEA are implemented by using different combinations of the three operators, namely, FCWNEA_{OP₁} using OP₁, FCWNEA_{OP₂} using OP₂, FCWNEA_{OP₃} using OP₃, FCWNEA_{OP₁+OP₂} using OP₁ + OP₂, and FCWNEA using OP₁ + OP₂ + OP₃. Five different versions of FCWNEA are compared on Mv_DTLZ 1,4 and Mv_UF 1-2.

The empirical results are shown in Table 4. Two observations can be made.

Table 4

Average and standard deviation of HV metric values obtained by five versions of FCWNEA with different combinations of operators on Mv_DTLZ1,4 and Mv_UF1,2 problems. The bolded values in the dark gray and gray backgrounds indicate the best and second-best results among the different versions of FCWNEA based on the mean metric value. '+', '-' and '=' indicate that the result is statistically better, worse and similar to that of FCWNEA, respectively.

Problem	FCWNEA_1	FCWNEA_2	FCWNEA_3	FCWNEA_1+2	FCWNEA
Mv_DTLZ1	8.2341e-1 (4.08e-2) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	8.2920e-1 (1.26e-2) -	8.3832e-1 (1.64e-5)
Mv_DTLZ4	4.9446e-1 (2.87e-2) -	3.5209e-1 (3.48e-2) -	2.2777e-1 (1.48e-1) -	4.0827e-1 (2.93e-2) -	4.9876e-1 (3.26e-2)
Mv_UF1	6.9826e-1 (1.05e-2) -	6.7929e-1 (1.40e-2) -	5.9403e-1 (6.73e-3) -	7.0874e-1 (4.68e-3) +	7.0586e-1 (9.98e-3)
Mv_UF2	7.0619e-1 (2.73e-3) -	7.0001e-1 (3.88e-3) -	6.6174e-1 (8.93e-3) -	7.0493e-1 (2.72e-3) -	7.0653e-1 (2.08e-3)

**Fig. 9.** The final solution obtained by FCWNEA and NSGA-III on Mv_DTLZ4.

- The OP_1 operator is the most important. The version with OP_1 is significantly better than the version with OP_2 or OP_3 . In particular, for the Mv_DTLZ1 problem, the algorithm using OP_2 or OP_3 does not even converge to near PF.
- The performance of FCWNEA can be improved when OP_1 is used in combination with other operators. From Table 4, it can be concluded that the performance of different combinations of operators is ordered as $OP_1 < OP_1 + OP_2 < OP_1 + OP_2 + OP_3$.

5.5. The effect of the node-value adjustment strategy

As described in Section 3.4, FCWNEA searches continuous variables by dynamically adjusting the node values. To demonstrate the effectiveness of the treatment of continuous variables, we compare FCWNEA with and without the node-value adjustment strategy on the Mv_DTLZ4 problem. Mv_DTLZ4 is a biased problem, that is, when the population is uniformly distributed in the target space, it is clustered in a narrow region in the decision space. The strategy's effectiveness involving the dynamic adjustment of node values for continuous problems can be tested using that property. Initially, the node values are uniformly distributed throughout the space, and as it evolves, the node values of the continuous variables are adaptively adjusted to be distributed in the narrow region where the optimal PS is located.

The experimental results are shown in Fig. 9. FCWNEA with a dynamic adjustment strategy can cover the PF better, as shown in Fig. 9 (b), while FCWNEA without a dynamic adjustment strategy can only converge with a few points. In the FCWN model, the values of the continuous variables are selected from a finite set of values that have already been determined. Without a dynamic adjustment strategy for node values, the search positions of continuous variables are fixed, and the search granularity is large, being 1, which means that the search precision is also small. As shown in Fig. 9 (a), the interval between the values in the 1-st and 2-nd variables taken by the population is larger. However, using the strategy of dynamic adjustment of node values, as shown in Fig. 9 (c), the values of the population in the 1-st and 2-nd variables are concentrated in the range of [48.5, 50], which means that the search accuracy in the interval [48.5, 50] is much higher than the other regions.

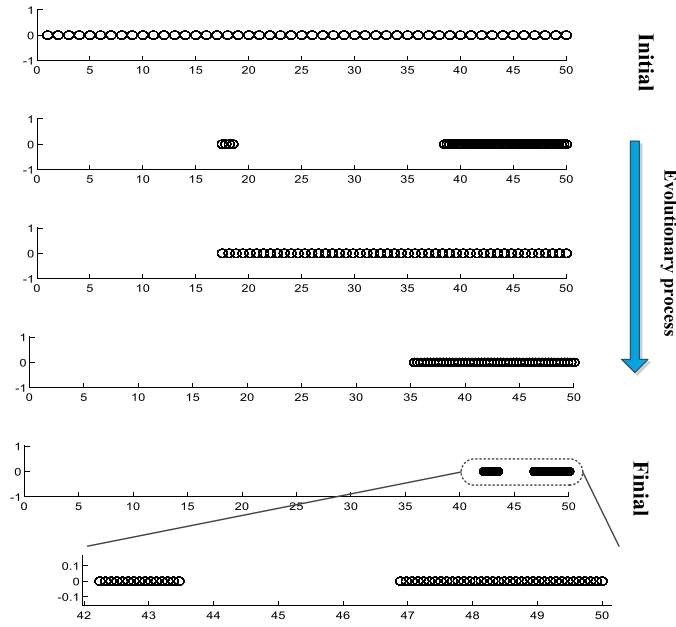


Fig. 10. The change processes of the node value in the evolutionary process.

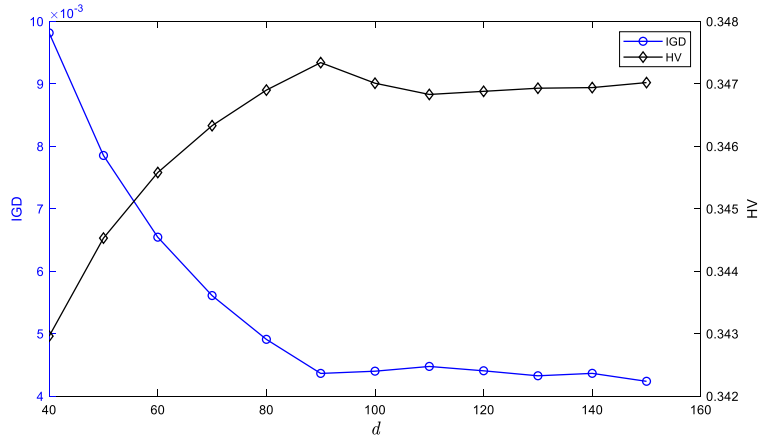


Fig. 11. The optimization results of FCWNEA on Mv_DTLZ4 with different d values varying from 40 to 140.

During the optimization process, the dynamic change process of the node values taken by the node layer of the 1-st variable is shown in Fig. 10. The node values are uniform at initialization and are determined by the variable value domain and the number of nodes in the network layer. Then, the node values are gradually and dynamically adjusted to promising regions, the search granularity decreases, and the search accuracy increases. Additionally, for continuous variables, the parameter d significantly determines the number of discrete values that the continuous variable can take and the number of nodes in the continuous variable layer. To investigate the influence of the parameter d , we apply FCWNEA to optimize Mv_DTLZ4 with different values of d . The values considered here range from 40 to 140 with an increment of 10. The optimization results are assessed using the HV and IGD metrics and depicted in Fig. 11. The results indicate that the proposed algorithm attains the optimal performance when $d = 90$, which is very close to the population size.

5.6. Performance on real-world application

To demonstrate the effectiveness of FCWNEA in tackling real-world problems, we apply FCWNEA to a configuration optimization problem of an integrated energy system (IES) that combines combined cooling, heating, and power (CCHP) with solar and wind power (CCHP-SW) and compare its performance with other state-of-the-art algorithms. The CCHP-SW system comprises various kinds of energy generation, conversion, and storage devices interconnected by multiple energy flows. These devices' optimal configurations are essential for minimizing energy consumption, reducing costs, and lowering carbon emissions. Therefore, we aim to optimize

Table 5

Average and standard deviation of the HV metric values obtained by the compared algorithms. The Wilcoxon-Ranksum non-parametric test at the 95% confidence level is performed. '+', '-' and '=' indicate that the result is significantly better, significantly worse and statistically similar to that of FCWNEA. The best HV value among these algorithms is highlighted in bold.

N_type	FCWNEA	NSGA-III	MVMO_EDA	MOEAD	PREA
5	5.5111e-1 (2.56e-3)	5.3577e-1 (1.40e-2) -	4.1879e-1(1.41-2) -	3.6235e-1 (4.22e-3) -	5.4989e-1 (1.76e-2) -
8	5.5557e-1 (2.34e-3)	5.4789e-1 (1.29e-2) -	4.3447e-1(2.62e-2) -	3.5313e-1 (1.10e-2) -	5.5169e-1 (1.34e-2) -
10	5.6208e-1 (3.95e-3)	5.6031e-1 (1.26e-2) -	4.6084e-1(2.47e-2) -	3.6855e-1 (8.15e-3) -	5.5616e-1 (9.34e-4) -

the type and the number of the CCHP-SW components as well as some key parameters, such as the angle of the photovoltaic panel and the height of the wind turbines. The configuration optimization of the CCHP-SW system is formulated as a mixed-variable multi-objective problem. This problem involves different types of variables, including categorical variables (such as equipment type), integer variables (such as the number of equipment), and real variables (such as the height of the wind turbine). Moreover, it encompasses multiple conflicting objectives, including the annualized system cost, the percentage of renewable energy, the load supply deficiency rate, and the system's independence. More details about the configuration optimization of the IES can be found in the appendix. Table 5 lists the means and standard deviations in terms of the HV values of the configurations set, obtained by the proposed FCWNEA and the other algorithms. Compared to other approaches, FCWNEA can achieve better mean results than other algorithms, which means that FCWNEA can provide decision makers with better performance and more diverse configuration options.

6. Conclusion

How to handle mixed-variable multi-objective optimization problems has not been well studied. Integer and categorical variables make the decision space discrete. In particular, categorical variables break the nearest-neighbor relation of variable values. The existing widely used operational operators for continuous multi-objective optimization problems, such as SBX, PM, and DE, can apply to integer and categorical variables with simple modifications, but they do not work well. Thus, in this paper, a probability distribution model based on a fully connected weight network is designed, named the FCWN, in which the decision space is characterized by a fully connected network, and the distribution is estimated by the weights of connections, node visits, and network layer activity. A mixed-variable multi-objective evolutionary algorithm based on the model FCWN is constructed, named FCWNEA. The FCWN model can be seen as a statistical and evaluation model of the entire search history. Three genetic reproduction operators based on the model FCWN are designed to generate offspring solutions. The variable value dynamic adjustment strategy is proposed to accommodate the search for continuous variables. Experimental studies have shown that FCWNEA performs well on mixed-variable multi-objective optimization problems.

Future research topics along this line include the following.

- Further refinement of the algorithmic framework. The FCWN model proposed in this paper is a preliminary framework with more room for improvement.
- Incorporating other environmental selection strategies into the FCWNEA to improve its global search capability for mixed-variable multi-objective problems.
- Designing novel model update mechanisms to improve the learning ability of historical search knowledge. Designing more efficient gene operators to generate offspring solutions in promising regions more efficiently.
- Further investigating the parameter settings in FCWNEA, such as the search granularity of continuous variables and the proportion of different operators in different search stages.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ins.2023.120053>.

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